

1 **An Evaluation Ladder for Regional Weeks-to-Years Probabilities of**
2 **Temperature, Precipitation, and Runoff**

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6 ABSTRACT: Reservoir operations, hydropower, and cooling water depend on regional tempera-
7 ture, precipitation, and runoff at leads of weeks to years. The atmospheric initial condition is largely
8 lost at these leads, and the useful quantity is a shift in the probability of a hot, cold, wet, or dry
9 period. Earth system foundation models are commonly accepted on the root mean square error of
10 the forecast field. A squared error is minimized by the mean field, so a model can train successfully
11 and still leave event probabilities at their historical rate. This paper specifies an evaluation ladder
12 for those probabilities before training. Each rung changes one design axis: the target, the season,
13 the lead, the variable, or the forecast object. Training years alone fix the category boundaries
14 and the historical class frequencies. A claim is kept only when the Brier skill score is positive
15 on every held-out sample. The same events are scored over the Tennessee Valley in ERA5 and
16 in E3SM seasonal-to-decadal hindcasts initialized on 1 May and 1 November. In ERA5, winter
17 precipitation probabilities remain skillful to a lead of two seasons, and one-week air temperature
18 and surface runoff probabilities are skillful when the issue-week atmosphere is included. In the
19 hindcasts, one-week temperature skill recurs for November initializations; week-ahead runoff and
20 winter precipitation do not. The claim tables are the milestone control for a later model, on the
21 leads this sample defends. A row marked yes is an advance only when that model beats the logistic
22 with the same issue-time information, on the same edges and the same held-out splits, in the dataset
23 where it was trained. A row marked thin or no uses the historical class rate as its floor. Beating
24 that rate is a separate result from beating the logistic. The scores are point estimates. A pass in
25 one dataset is not a pass in the other, and no row covers a lead of a year.

26 SIGNIFICANCE STATEMENT: This paper proposes evaluation criteria for weeks-to-years pre-
27 diction of water and heat for energy systems. At those leads, reservoir, hydropower, and cooling-
28 water decisions depend on whether a region will be unusually hot, cold, wet, or dry, and determin-
29 istic scores of the forecast field can favor the historical pattern without shifting those probabilities.
30 The criteria form an evaluation ladder: one element of the forecast design changes at a time, and
31 a result is retained only when the Brier skill score is positive on every held-out sample against
32 a historical rate fixed on training years. Applied to ERA5 and to the E3SM seasonal-to-decadal
33 hindcast campaign over the Tennessee Valley, the ladder is the milestone control for a later foun-
34 dation model: an advance only on a row marked yes, against the logistic with the same issue-time
35 information, and only in the dataset where that model is trained.

36 1. Introduction

37 Weeks-to-years prediction of water and heat is the lead at which an energy system plans. In the
38 Tennessee Valley, reservoir operations, hydropower scheduling, and the cooling water available to
39 thermal generation depend on rainfall, streamflow, and river temperature over the coming season
40 and the coming year. Fuel procurement and maintenance are set on that same horizon. The decisions
41 sit past the reach of medium-range weather guidance and short of a climate projection (Vitart et al.
42 2017; Saha et al. 2014). The products that do cover this range, including operational seasonal
43 forecasts out to about a year, are issued largely as anomalies from a historical weather pattern (Saha
44 et al. 2014; Johnson et al. 2019). When hot, dry, and high-flow seasons depart from that pattern,
45 the quantity an operator can use is a change in the probability of the event, relative to the rate the
46 event already has. Near-surface temperature, precipitation, and runoff are the variables through
47 which this paper poses that question for the valley.

48 The prediction is hard because the information that makes a short forecast skillful does not
49 survive to these leads, and the climate mean is not yet a seasonal decision. The atmospheric
50 initial condition is largely lost within weeks (Vitart et al. 2017). What remains is memory in
51 the coupled system, including the El Niño–Southern Oscillation and its effect on North American
52 precipitation, and that memory appears as a shift in the probability of a regional class (Ropelewski
53 and Halpert 1986; Hong et al. 2025). Physics-based systems built for the interval therefore use
54 initialized hindcasts of about one to two years and score seasonal anomalies after a lead-dependent

55 model climatology is removed. The Climate Forecast System, SEAS5, and the CESM Seasonal-to-
56 Multiyear Large Ensemble follow this design. The last of these starts two-year hindcasts four times
57 a year because seasonal protocols stop too soon, and decadal protocols start too seldom, to sample
58 the interval (Saha et al. 2014; Johnson et al. 2019; Yeager et al. 2022). Regional skill in those
59 systems remains limited, and the guidance that reaches an operator stays close to the historical
60 pattern.

61 Foundation models are now trained on reanalysis and on coupled-model output to learn a predic-
62 tive operator for Earth-system fields. GraphCast, Pangu-Weather, and ClimaX are accepted on the
63 root-mean-square error and the anomaly correlation of atmospheric maps at leads of hours to about
64 ten days (Lam et al. 2023; Bi et al. 2023; Nguyen et al. 2023). ORBIT and ORBIT-2 carry that class
65 of model toward Earth-system predictability and toward weather and climate downscaling (Wang
66 et al. 2024, 2025). Those scores ask whether the predicted map is close to the analysis. A squared
67 error is minimized by the mean field, so a model can train successfully by reproducing the historical
68 pattern and still leave the probability of an unusual heat or water event at its historical rate. A loss
69 on the continuous amount averages that shift away, and additional capacity can fit weather noise
70 while missing a shift that a small probability model still detects. Seasonal and subseasonal systems
71 already score categorical events with the Brier skill score (Vitart et al. 2017; Kirtman et al. 2014).
72 That score has not been the acceptance test for a foundation model of regional water and heat at
73 weeks-to-years leads.

74 This paper fixes that test before such a model is trained. The evaluation ladder changes one design
75 axis at a time: the target, the season, the lead, the variable, or the forecast object. Category edges
76 and the historical class rate are frozen on training years. The score is the Brier skill score (Brier
77 1950; Wilks 2011), and a result is retained only when the skill is positive on every pre-declared
78 held-out split. The same events are scored over the Tennessee Valley in ERA5 (Hersbach et al. 2020)
79 and in the E3SM seasonal-to-decadal hindcast campaign, initialized on 1 May and 1 November
80 for about 24 months (Golaz et al. 2022). A pass in the reanalysis is evidence about the observed
81 record. A pass in the hindcasts is evidence about the model’s own categories. One result is not
82 copied onto the other. Section g is the rule a later model is scored under. The leads it can defend
83 are the rows of Table 5. A lead of a year is not among them.

84 2. Evaluation Ladder

85 a. One axis at a time

86 Each rung changes one of five things: the target type (a continuous amount or a category), the
87 season, the lead, the variable, or the forecast object (a regional mean or a map). Predictors are a
88 menu declared before the rung is scored. A failure under that menu is a result for the menu. It
89 is not a license to search for a new predictor on the same test sample and then quote the original
90 experiment.

91 b. The event

92 The scored object is a regional-mean category. For a water variable the classes are dry, near,
93 and wet. For temperature they are cool, near, and hot. The two tercile edges, and the historical
94 frequency of each class, are estimated on training time only and then frozen. Validation and test
95 samples are labeled with those edges. They do not move the edges.

96 c. The score

97 Let $p_{k,c}$ be the forecast probability of class c at occasion k , and let $o_{k,c}$ be 1 if that class occurs
98 and 0 otherwise. The Brier score and its skill against the training class frequencies $\bar{o}_c$ are

$$\text{BS} = \frac{1}{N} \sum_{k=1}^N \sum_c (p_{k,c} - o_{k,c})^2, \quad (1)$$

$$\text{BSS} = 1 - \text{BS}/\text{BS}_{\text{ref}},$$

99 where BS_{ref} uses $p_{k,c} = \bar{o}_c$ (Brier 1950; Wilks 2011). A positive skill score means the probabilities
100 beat that historical rate. Hit rate and the Heidke skill score (Heidke 1926) are reported beside it. A
101 claim requires $\text{BSS} > 0$ on every pre-declared held-out split, for that variable and that lead. A single
102 season with a positive score does not replace the pooled score. Section g separates that clearance
103 from an advance, and it keeps a thin positive off the numeric floor. The probability model used to
104 test the ladder is logistic regression. It is large enough to express a shift in class probabilities and
105 small enough that a pass is unlikely to be memorized weather.

106 *d. The baseline is declared per dataset*

107 The historical rate in (1) is the training class frequency of the dataset being scored. It is the
108 baseline of that experiment. It is not assumed to be a 30-year climate normal. In the week and
109 month rungs of the reanalysis case, the years that define the edges are 2010–2014, because those
110 are the training years of the daily pack. A new dataset names its own training years before any
111 score is computed. Using one normal for both a reanalysis and a coupled-model ensemble is a
112 separate, named choice. It is not the default here.

113 *e. Two worlds, one protocol*

114 Observation samples and coupled-model samples may use different year lists, different ensemble
115 sizes, and different variable names. They use the same event definition, the same edge rule, the
116 same score, and the same pass rule. A pass in one world is not written into the other world's table.

117 *f. What a map may add*

118 A map is scored only after the regional-mean probability has cleared Section c. A pattern score
119 does not rescue a failed regional mean. A deterministic score of the map, such as its root-mean-
120 square error, is a different experiment. A smaller value of that score does not clear the probability
121 row. A higher regional-mean score than the logistic is an improvement on that probability. It does
122 not show that a spatial field contained information the regional predictors lacked. Section g is the
123 rule that uses this distinction.

124 *g. Milestone rule for a later model*

125 The ladder is frozen before a foundation model is trained. It is the control for the leads that have
126 been scored. An unscored lead, including a lead of a year, has no floor in this paper.

127 A row is yes, thin, or no. Yes requires Brier skill above zero on every pre-declared held-out split,
128 with a margin the claim table treats as one the sample can carry. Thin is a small positive Brier
129 skill, or a positive Brier skill with a negative Heidke score. No is every remaining row. Where a
130 rung also requires named season slices to be positive, a pooled score with a negative required slice
131 is no. A single season with a positive score does not replace the pooled score.

132 Clearance and advance are different results. Clearance is $BSS > 0$ against the historical rate in
133 (1) on every pre-declared split. An advance is available only on a yes row. It requires clearance,
134 and a Brier score lower than the matched logistic on those same splits. On a thin row and on a no
135 row, the floor is the historical rate. The recorded thin skill is kept in the table. It is not the number
136 a later model must beat.

137 The matched logistic is the pre-declared menu whose inputs the later model is allowed to use at
138 issue time. Indices alone, and indices plus the spatial mean of the issue-period atmosphere, are
139 different menus. A model that receives that atmosphere is scored against the atmosphere menu. A
140 model limited to the slow indices is scored against the indices-only menu. When the atmosphere
141 menu scores below the indices-only menu, the atmosphere menu is not the control. A menu chosen
142 by searching the same test sample after the rung is closed is not a control.

143 The same held-out rows fix the logistic and then receive the later model. The comparison is
144 paired and descriptive. This paper does not compute a block bootstrap, a permutation test, or a
145 minimum margin. A yes row is a closed control on that split. It is not a significance claim. Years
146 after the current test window lie outside the milestone.

147 The score is computed in the dataset the later model was trained on. An ERA5 yes row is not
148 an E3SM yes row. A model scored against the hindcast's own terciles is not a forecast of the
149 observed valley. The edges remain the training ruler of Section d. A named climate normal, such
150 as 1991–2020 or 1980–2009, is a new scorecard. Tercile classes are the event this milestone scores.
151 A reservoir or cooling-water threshold is a different event. Reliability of the issued probabilities is
152 not part of the milestone. Where inside the valley the event sits, against a climatological probability
153 map, is a separate experiment.

154 **3. Reanalysis Case**

155 *a. Region and series*

156 The case is the Tennessee Valley in ERA5 (Hersbach et al. 2020). Seasonal rungs use a monthly
157 index pack: autumn Pacific indices for winter precipitation, and autumn Atlantic and Arctic indices
158 for winter air temperature, with training seasons through 2007 and 15 test seasons in 2009–2023.
159 Week and month rungs use daily fields on the box 34.0° – 37.8° N, 90.5° – 81.0° W. The week split is
160 259 training weeks (2010–2014), 155 validation weeks (2015–2017), and 103 test weeks (2018–

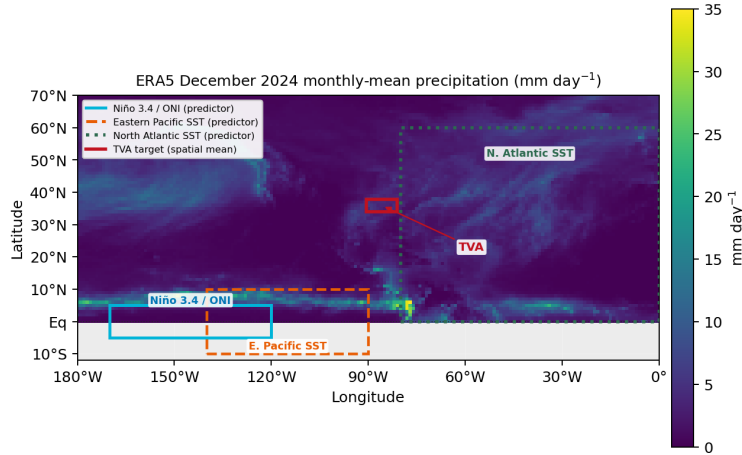


FIG. 1. Seasonal case geography. The red box is the Tennessee Valley mean. Pacific boxes supply the winter precipitation predictors. The North Atlantic box is the temperature-side sea-surface region. The field is one ERA5 precipitation month, shown only to locate the boxes.

2019). The month split holds out 2008 for validation and 2009–2023 for test. These two year lists are not one sample. Scored variables are 2 m air temperature, precipitation, and ERA5 surface runoff. Air temperature is not river temperature. Surface runoff is a reanalysis flux, not a stream gauge. Figure 1 shows the seasonal predictors and the valley whose mean is the target.

b. Rungs

Table 1 lists the axis each rung changed. The first rung asked whether indices forecast the monthly amount. Subsequent rungs replaced that target with categories, then moved the season, the lead, the variable, and finally the object from a regional mean to a map. At the week and month leads, a second logistic appends the spatial mean of the issue-period atmosphere to the indices. That second model tests whether the recent large-scale state carries probability information beyond the slow indices. It is still a regional-mean model.

c. Seasonal categories

All seasonal scores use 25 training seasons and 15 test seasons (2009–2023). There is no separate validation year in this protocol. Hit rate is the fraction of seasons in which the most likely class is correct. The majority-class hit for balanced terciles is about one third. Heidke skill corrects the hit

Rung	Axis	Question
Amounts	Target	Do indices forecast the monthly amount?
Winter categories	Target	Do autumn indices move winter odds?
Other seasons	Season	Does that menu work in spring, summer, and fall?
Longer winter lead	Lead	Do winter odds survive two and three seasons?
Maps	Object	Is a week-ahead rain map the product?
One week	Lead, variable	Do next-week odds beat the historical rate?
Two and three weeks	Lead	Do one-week odds survive another week or two?
Two and three months	Lead	Do water odds at those leads beat the historical rate?
Larger winter models	Capacity	Does a larger net beat the logistic on the same labels?

TABLE 1. One axis changes at each rung.

Target	Event	Hit	Heidke	BSS	Majority hit	Result
Winter	Precipitation	0.667	0.432	+0.091	0.333	Pass
Winter	Air temperature	0.400	0.167	+0.009	0.133	Pass, thin
Winter	Air temperature, same-season indices	0.133	-0.096	-0.129	0.133	Fail
Spring	Precipitation	0.333	-0.304	-0.207	—	Fail
Spring	Air temperature	0.133	-0.016	-0.122	—	Fail
Summer	Precipitation	0.400	0.075	-0.011	—	Fail
Summer	Air temperature	0.333	0.063	-0.091	—	Fail
Fall	Precipitation	0.400	-0.125	+0.016	—	Thin
Fall	Air temperature	0.400	-0.063	-0.295	—	Fail

TABLE 2. Seasonal logistic odds, one season ahead, 15 test seasons. Pass requires Brier skill above zero and either Heidke skill above zero or a hit rate above the majority class.

rate for those class frequencies (Heidke 1926). Table 2 applies one predictor menu to every season: autumn Pacific indices for precipitation (the Oceanic Niño Index and the Niño-3.4 region (NOAA Climate Prediction Center 2026; Barnston et al. 1997)) and autumn Atlantic and Arctic indices for temperature (Hurrell 1995; Thompson and Wallace 1998). Winter rows are the reference rung. The other rows change only the target season.

Winter precipitation is the strong row: about two winters in three, against a one-third historical rate. The wet/dry calls are uneven. The model rarely selects the dry class, so the skill is a shift toward wet in some winters and toward the middle class in others. The probability score is still the largest in the ladder. The same Pacific predictors, scored as a continuous winter precipitation anomaly rather than a category, are only marginal (skill against the mean +0.02, anomaly correlation 0.44). Continuous winter temperature fails (skill against the mean -0.04). Same-season Atlantic indices,

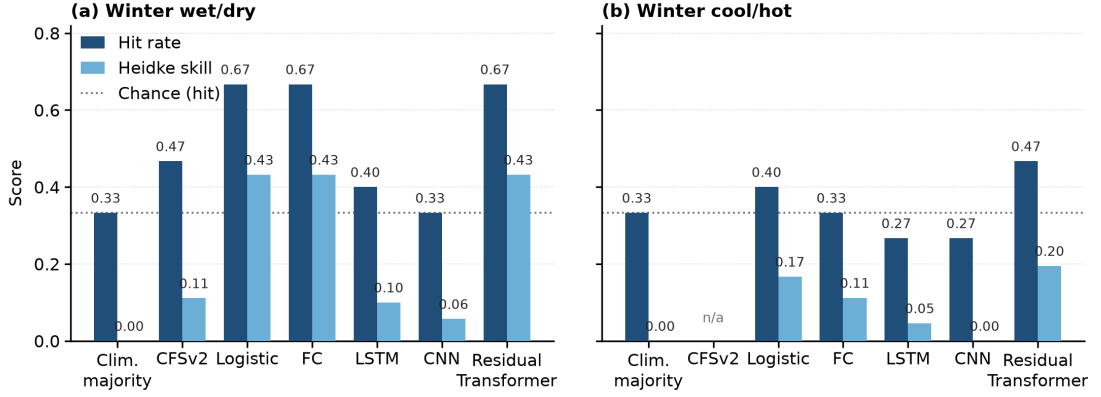


FIG. 2. Blocked test on 15 winters. (a) Winter precipitation categories, including October-initialized CFSv2 on the same ERA5 labels. (b) Winter temperature categories. CFSv2 was not scored on temperature. The dotted line is a hit rate of one third. Logistic is the reference. The residual Transformer matches its hit rate and Heidke skill on precipitation and does not classify more winters correctly.

which have no lead, also fail the winter temperature categories (BSS -0.129). A contemporaneous correlation and a category forecast at a lead are different results.

Spring and summer fail for both variables. Fall precipitation is the only non-winter positive probability score ($+0.016$), and its Heidke score is negative, so the claim is thin. Fall temperature fails.

On the same 15 winters, October-initialized CFSv2 ensemble-mean precipitation was mapped onto the valley box and labeled with the same ERA5 terciles (Saha et al. 2014; Kirtman et al. 2014). It is correct on 7 winters (hit 0.467, Heidke 0.111). The logistic is correct on 10 (hit 0.667, Heidke 0.432). CFSv2 sits above the one-third majority and below the logistic. The comparison is categorical: the ensemble mean was assigned a class, so CFSv2 has no Brier skill in this table. Figure 2 places that comparison next to the other models trained on the same labels. Figure 3 shows why the operational hit rate stays near the majority. The continuous October forecast is nearly flat through the test winters, while the ERA5 winter anomalies are not. The categorical panel is the score that matches an odds product.

The fully connected net copies the logistic hit rate and loses probability skill (0.051 against 0.091). The LSTM and the CNN fail the precipitation categories at this sample size. The residual Transformer keeps the logistic path and adds a constrained attention correction (Vaswani et al.

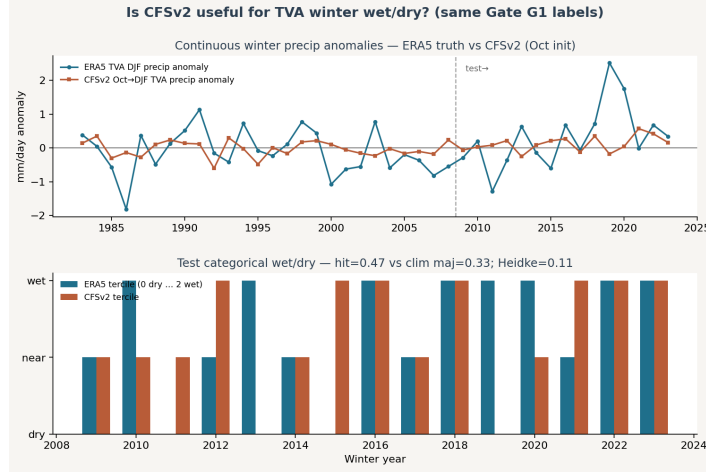


FIG. 3. October-initialized CFSv2 on the Tennessee Valley, scored against the same ERA5 winter precipitation labels as the logistic. Top: the forecast anomaly stays near zero while the reanalysis anomaly does not. Bottom: the resulting categories on the 15 test winters. CFSv2 is correct on 7 winters. The logistic is correct on 10. This is a local categorical check, not a global verification of CFSv2.

Event	Model	Hit	Heidke	BSS
Precip.	CFSv2 Oct→DJF	0.467	0.111	—
Precip.	Logistic	0.667	0.432	+0.091
Precip.	Fully connected	0.667	0.432	+0.051
Precip.	LSTM	0.400	0.100	−0.016
Precip.	CNN	0.333	0.057	−0.038
Precip.	Residual Transformer	0.667	0.432	+0.140
Temp.	Logistic	0.400	0.167	+0.009
Temp.	Residual Transformer	0.467	0.195	+0.050

TABLE 3. Same 15 winter test seasons. CFSv2 is an ensemble-mean class, so its Brier skill is not defined here.

2017). On precipitation it matches the logistic hit and Heidke and raises Brier skill from 0.091 to 0.140. On temperature it is correct on one more winter (0.467 against 0.400). It does not replace the logistic as the reference. At the two-season precipitation lead the same style of model fell below the logistic, which is the comparison used in Table 4.

Table 4 keeps winter as the target and moves the issue earlier. A lead of one season reproduces the winter rows of Table 2.

Precipitation skill falls from +0.091 at one season to +0.024 at two seasons and to −0.230 at three (Table 4). Temperature does not survive past one season. The logistic remains the reference

Lead	Event	Hit	Heidke	BSS
1 season	Precipitation	0.667	0.432	+0.091
1 season	Air temperature	0.400	0.167	+0.009
2 seasons	Precipitation	0.400	0.135	+0.024
2 seasons	Air temperature	0.267	-0.071	-0.084
3 seasons	Precipitation	0.133	-0.121	-0.230
3 seasons	Air temperature	0.200	-0.208	-0.056

TABLE 4. Winter odds at longer leads. Issue season is one, two, or three seasons before winter. Same 15 test winters.

Event	Lead	Held-out BSS	Claim
Winter precipitation	1 season	+0.091	yes
Winter air temperature	1 season	+0.009	thin
Fall precipitation	1 season	+0.016	thin
Winter precipitation	2 seasons	+0.024	thin
Week air temperature	1 week	+0.018/+0.032	yes
Week runoff	1 week	+0.018/+0.052	yes
Week runoff	2 weeks	+0.033/+0.014	thin
Week precipitation; month water	1 week; 2–3 months	Tables 6–8	no
Other two- and three-week rows	2–3 weeks	Table 7	no

TABLE 5. Claims retained from the reanalysis case. Thin means a small positive Brier skill, or a positive Brier skill with a negative Heidke score. Under Section g a thin row is not a numeric floor. Its floor is the historical rate. A yes row is the row a later model must beat.

at both leads, including after the model comparison in Table 3. Table 5 collects the claim that follows from Tables 2 and 4.

d. One week ahead

The week rung uses the next ISO week after the issue Sunday. Training, validation, and test contain 259, 155, and 103 weeks. A pass requires Brier skill above zero on both the validation weeks and the test weeks. Table 6 separates indices at the issue Sunday from the same indices plus the spatial mean of that week’s 1° atmosphere (temperature, precipitation, and sea-level pressure) and the month of the year. Summer and winter columns are slices of the test set. They are reported so a pooled pass can be read by season. They are not a substitute for the pooled pass.

Two rows pass. Air temperature with the issue-week atmosphere is positive on validation (+0.018) and on test (+0.032). The test skill is in summer (+0.212). Test winter is negative

Variable	Predictors	Val	Test	Test summer	Test winter
Air temperature	Indices	−0.114	−0.040	+0.063	−0.101
Air temperature	Indices + atmosphere	+0.018	+0.032	+0.212	−0.119
Precipitation	Indices	−0.003	+0.031	−0.007	+0.066
Precipitation	Indices + atmosphere	−0.003	+0.066	+0.031	+0.176
Surface runoff	Indices	−0.005	+0.024	−0.005	+0.053
Surface runoff	Indices + atmosphere	+0.018	+0.052	+0.077	+0.078

TABLE 6. One-week Brier skill for the Tennessee Valley mean. Indices alone fail every variable. The atmosphere column adds the issue-week spatial mean.

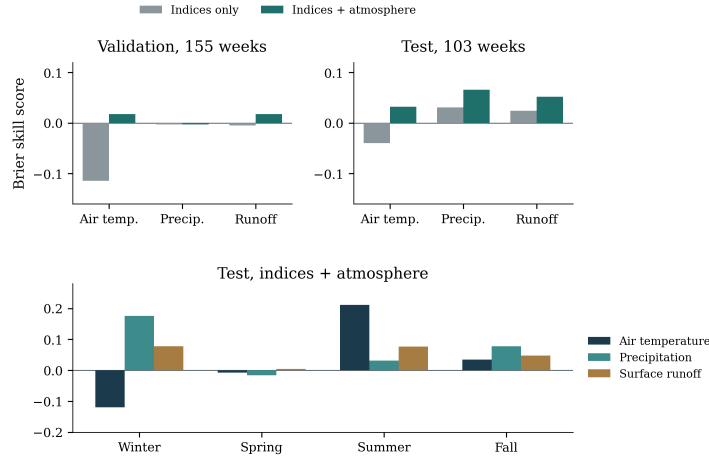


FIG. 4. One-week Brier skill for the valley mean. Left and center: indices alone against indices plus the issue week's atmosphere. A pass requires both of those panels to be above zero. Right: the passing predictor set, split by season on the test weeks.

(−0.119), so this is not a winter temperature forecast. Surface runoff with the same predictors is positive on validation (+0.018) and on test (+0.052), and the test summer and test winter slices are both positive (+0.077 and +0.078). Hit rates on the test weeks for these two passing models are 0.47 for temperature and 0.45 for runoff. Precipitation with the atmosphere reaches +0.066 on the test weeks, including +0.176 in test winter, and remains negative on validation (−0.003). The pass rule therefore rejects week-ahead precipitation. Figure 4 is that comparison. The right-hand season panel is not the pass rule. It shows where the test skill sits: temperature skill is in summer, and winter temperature is negative.

The same week list, scored as precipitation amount maps rather than as odds, loses to the historical field: root-mean-square error 0.64 for the map model and 0.53 for the historical map.

Lead	Predictors	Val	Test	Test summer	Test winter	Pass
<i>Air temperature</i>						
+2 weeks	Indices	−0.106	−0.037	+0.052	−0.103	no
+2 weeks	Indices + atmosphere	−0.060	−0.026	+0.166	−0.158	no
+3 weeks	Indices	−0.097	−0.037	+0.051	−0.086	no
+3 weeks	Indices + atmosphere	−0.055	−0.043	+0.147	−0.190	no
<i>Precipitation</i>						
+2 weeks	Indices	−0.005	+0.015	−0.000	+0.038	no
+2 weeks	Indices + atmosphere	−0.003	+0.009	+0.013	+0.106	no
+3 weeks	Indices	−0.019	−0.015	+0.004	−0.025	no
+3 weeks	Indices + atmosphere	−0.038	−0.025	+0.011	+0.045	no
<i>Surface runoff</i>						
+2 weeks	Indices	−0.007	+0.021	−0.008	+0.035	no
+2 weeks	Indices + atmosphere	+0.033	+0.014	+0.055	+0.030	thin
+3 weeks	Indices	−0.010	+0.009	−0.017	+0.016	no
+3 weeks	Indices + atmosphere	−0.008	−0.040	+0.035	−0.019	no

TABLE 7. Two- and three-week Brier skill for the Tennessee Valley mean. Each lead is a separate logistic with its own terciles. One row passes.

e. Two and three weeks ahead

The longer week leads use the same issue weeks and the same two logistics. The target is the valley mean two ISO weeks ahead, and, in a separate fit, three ISO weeks ahead. Issue-time predictors stay on the issue week. Tercile edges are refit on that lead’s training weeks. A row is dropped when the target week is not in the same split as the issue week, so lead +2 has 258, 154, and 102 weeks and lead +3 has 257, 153, and 101. Table 7 is the same pass rule as Table 6.

One row clears the bar. Surface runoff at two weeks, with the issue-week atmosphere, is positive on validation (+0.033) and on test (+0.014). Test summer and test winter are both positive (+0.055 and +0.030). Test fall is −0.030 and validation fall is −0.011. The hit rate on the 102 test weeks is 0.40 and the Heidke score is +0.12. The skill is smaller than the one-week runoff row, and Table 5 marks it thin. Temperature at both longer leads keeps a positive test-summer slice and a negative test-winter slice, and the pooled scores are negative. Two-week precipitation reaches +0.106 in test winter and remains negative on validation (−0.003). At three weeks, no variable is positive on both splits.

Lead	Model	Val	Test	Test summer	Test winter	Pass
<i>Precipitation</i>						
+2 months	Indices	−0.005	−0.006	−0.008	−0.015	no
+2 months	Indices + atmosphere	−0.055	−0.031	−0.028	−0.032	no
+3 months	Indices	+0.021	−0.007	+0.001	−0.009	no
+3 months	Indices + atmosphere	−0.022	−0.034	−0.019	−0.014	no
<i>Surface runoff</i>						
+2 months	Indices	+0.055	−0.004	−0.023	+0.006	no
+2 months	Indices + atmosphere	−0.056	−0.001	+0.031	+0.032	no
+3 months	Indices	+0.033	+0.002	−0.006	−0.001	no
+3 months	Indices + atmosphere	−0.162	−0.049	−0.009	+0.009	no

TABLE 8. Month-ahead Brier skill. Each lead is a separate logistic. The atmosphere column adds the issue-month spatial mean and made every test score worse.

f. Two and three months ahead

The month rung uses every issue month from August 1982 through January 2023 (486 months). Training runs through December 2007 (305 months), validation is calendar 2008 (12 months), and the test is January 2009 through January 2023 (169 months). Leads of two and three months are fit separately, with separate tercile edges. Table 8 is Brier skill for precipitation and surface runoff. Persistence, a hard bet on the class of the issue month, is worse than the historical rate throughout this table, often with skill near -1 . It is not a useful null at these leads.

No cell clears the pass rule once summer and winter are both required to be positive. Three-month runoff with indices alone is the only pooled test score above zero, at $+0.002$, after a positive validation score of $+0.033$. Its test summer slice is -0.006 and its test winter slice is -0.001 . The pooled value is a residual across seasons, not a winter signal and not a summer signal. Section g marks that row no: a pooled score with a negative required season slice is not a claim. It is not entered in Table 5. The atmospheric mean lowers the test skill in every row, including that one. On the indices-only rows in Table 8, the pooled three-month runoff score is the only test value above zero, and its summer and winter slices are not.

g. What the failures fix in the score

The opening amount model used this month calendar and a masked squared error on standardized anomalies at leads of one to three months together. It scored near the long-term average and stopped at the first epoch. That run is why Tables 2–8 score categories rather than amounts. Same-day

maps can be trained on this archive, so a map failure at a lead is not a pipeline failure. Same-day surface runoff is readable from the atmosphere. Top-layer soil moisture is not, and soil moisture was left off the probability gate. Week-ahead rain maps and week-ahead rain probabilities fail together. On the winter labels, extra capacity raised the one-season probability score once and lost to the logistic at two seasons. The residual Transformer raised winter-precipitation Brier skill from +0.091 to +0.140 while matching the logistic hit rate and Heidke score. That gain is a change in the probabilities on the same class calls. It is not evidence that a spatial field contained information the regional predictors lacked. The event definition, not the size of the network, is what separates the rows that pass from the rows that do not.

4. Using the Ladder Again

a. A later model in the same reanalysis

A later model trained on these fields is reduced to Table 5 under Section g. The reduction uses the same issue times, the same regional mean, and the same frozen edges. The yes rows are winter precipitation at a lead of one season, one-week air temperature with the issue-week atmosphere, and one-week surface runoff with that same atmosphere. On those rows an advance must beat the matched logistic. The one-week temperature row is a pooled pass. Its test-winter slice is negative, so it is not a winter temperature forecast. Winter air temperature, fall precipitation, two-season winter precipitation, and two-week runoff are thin. Their floor is the historical rate in (1). Temperature and precipitation at two and three weeks, precipitation at one week, and every month-ahead water row are no. Their floor is the same historical rate. The week-ahead rows and the seasonal rows are different events. Passing one does not pass the other. A weekly reference map that the model displays, including a multi-decade week-of-year mean, is not the tercile ruler of Section d. Scoring against a newly chosen normal is a new case.

b. Coupled-model case

The second case asks whether the claims in Table 5 still shift categories in the E3SM seasonal-to-decadal (S2D) hindcast campaign. Starts are 1 May and 1 November in 1980–2003 and 2014–2018, under one model version. Training starts are 1980–1995, validation is 1996–2003, and the test is 2014–2018, including 2015. Each row is the mean of members EN00–EN02. The members share

Event	Lead	E3SM BSS	Repeats
Air temp.	1 week	+0.102/+0.131	Nov. only
Precip.	1 week	-0.017/-0.129	no
Runoff	1 week	-0.196/-0.628	no
Runoff	2 weeks	-0.027/-0.006	no
Precip.	2-3 mo	negative	no
Precip.	Nov→DJF	-0.059/+0.074	no
Air temp.	Nov→DJF	-0.043/-0.114	no

TABLE 9. E3SM skill on the reanalysis claims. BSS is validation/test. A repeat requires both above zero. The test has 10 starts for a week or month and 5 winters for a season.

the observed index, so they are not three samples. Terciles are refit on the training starts, separately for May and November. Week and month rungs then have 29 training starts, 16 validation starts, and 10 test starts. A seasonal rung has about 15 training starts and 5 test starts.

Predictors that must be known before the start are the two preceding months: September and October for a November start, March and April for a May start. The November value is not used, because that monthly index includes the rest of November. Pacific predictors are the Oceanic Niño Index, Niño-3.4, and the Niño-3.4 box anomaly. Temperature predictors at seasonal leads are the NAO and the AO. The week-ahead atmosphere is days 1–7 of the forecast. Temperature is 2 m air temperature. Precipitation is total precipitation. Runoff is total liquid runoff. Daily land history does not store the surface-runoff flux used in the reanalysis, so the runoff rows are a related event, not the same field.

Table 9 scores the reanalysis claims on that sample. One-week temperature repeats when the issue week is included (validation +0.102, test +0.131). Fit on November starts alone, the same row is +0.129 and +0.200. Fit on May starts alone, validation is -0.493, so the repeat is a November result. One-week precipitation still fails. One-week and two-week runoff, which pass in ERA5, do not. November-to-winter precipitation from the September–October Pacific indices does not, nor does November-to-winter temperature from the September–October NAO and AO. Precipitation at two and three months remains below the historical rate.

The written rule also marks cells that ERA5 did not pass. The largest is three-week runoff with the issue-week atmosphere (+0.253/+0.199). Two-week precipitation and runoff from the Pacific indices alone, runoff at two months from those indices, and a few seasonal runoff rows also clear both splits. November-to-winter runoff does so by +0.004 on validation. May-to-next-winter

345 temperature from the March–April NAO and AO does so by +0.020 on five winters. Those cells
346 are recorded. With 5 or 10 test starts they are not given the weight of the reanalysis claims. Under
347 Section g the hindcast yes row is one-week temperature for November initializations. The ERA5
348 yes rows for runoff and for winter precipitation do not transfer. A cell that clears both splits only
349 in the hindcast is not a yes row.

350 **5. Limitations**

351 Fifteen test seasons can support a research gate. They cannot support an operational hit-rate
352 claim. The winter precipitation model rarely calls the dry class, so the probability score is the more
353 stable summary. Validation at the month lead is a single year. The week split and the seasonal split
354 cover different years, and a sentence that merges them is not supported. The 2010–2014 terciles
355 are the training ruler of the week and month rungs, not a named climate normal. A failed season
356 is a result for the pre-declared index menu. The ensemble test is 10 starts for a week or a month
357 and 5 starts for a season, so an ensemble pass is thinner than a reanalysis pass. Ensemble runoff is
358 total liquid runoff, not the reanalysis surface flux. The ensemble score is against the model’s own
359 terciles, not against the observed valley. The CFSv2 comparison is a Tennessee Valley categorical
360 check on 15 winters (Barnston et al. 2012). It is not a global verification of that forecast system.
361 No block bootstrap is reported, so a thin positive is not a margin a later model can be required
362 to exceed. The held-out rows that fixed the logistic are the rows that will receive the later model.
363 That comparison does not use a fresh sample. Equal terciles are not a reservoir or cooling-water
364 threshold, and the reliability of the issued probabilities is not scored. No lead of a year has a score,
365 so the milestone does not cover that lead.

366 **6. Conclusion**

367 Weeks-to-years event probabilities are a hard forecast problem because a deterministic score of
368 the field, such as the root-mean-square error, can be minimized by the reference field. The ladder
369 isolates the alternative quantity: a shift of category probabilities above a historical rate that was
370 frozen before the test. In ERA5 that filter keeps winter precipitation odds out to two seasons,
371 one-week air-temperature and surface-runoff odds when the issue week’s atmosphere is included,
372 and a thinner two-week runoff odds from those same predictors. In the S2D hindcasts the same rule

373 keeps one-week temperature odds for November initializations and does not keep the week-ahead
374 runoff odds or the winter precipitation odds. A later model is an advance only on a yes row, only
375 against the logistic with the same issue-time information, and only in the dataset where it was
376 trained. A thin row and a no row keep the historical rate as their floor. Clearance of that rate is not
377 the advance. An unscored lead, including a lead of a year, is outside the milestone.

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388 *Availability statement.* The seasonal runs use the monthly AI-ready pack archived at <https://doi.org/10.13139/ORNLNCCS/3409936>. The E3SM seasonal-to-decadal AI-ready sample is
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391

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